Combinatorics

Rosen 8th ed., §§ 6.1-6.6,

6 Counting

- 6.1 The Basic of Counting
- **6.2 The Pigeonhole Principle**
- 6.3 Permutations and Combinations
- 6.4 Binomial Coefficients
- **6.5 Generalized Permutations and Combinations**
- 6.6 Generating Permutations and Combinations

Combinatorics

- The study of the number of ways to put things together into various combinations.
- *E.g.* In a contest entered by 100 people,
  - how many different top-10 outcomes could occur?
- *E.g.* If a password is 6-8 letters and/or digits,
  - how many passwords can there be?

Sum and Product Rules (§ 6.1)

- Let *m* be the number of ways to do task 1 and *n* the number of ways to do task 2,
  - with each number independent of how the other task is done,
  - and also assume that no way to do task 1 simultaneously also accomplishes task 2.
- Then, we have the following rules:
  - The *sum rule*: The task "do either task 1 or task 2, but not both" can be done in $m+n$ ways.
  - The *product rule*: The task "do both task 1 and task 2" can be done in mn ways.

Set Theoretic Version

- If A is the set of ways to do task 1, and B the set of ways to do task 2, and if A and B are disjoint, then:
  - The ways to do either task 1 or 2 are $A \cup B$, and $|A \cup B| = |A| + |B|$
  - The ways to do both task 1 and 2 can be represented as $A \times B$, and $|A \times B| = |A| \cdot |B|$

Example 1 ,2

- A new company with just two employees , Sanchez and Patel, rents a floor of a building with 12 offices. How many ways are there to assign different offices to these two employees?
- The chairs of an auditorium are to be labeled with an uppercase English letter followed by a positive integer not exceeding 100. What is the largest number of chairs that can be labeled differently?

Example 6 ,7

- Counting functions:
- How many functions are there from a set with *m* elements to a set with *n* elements?
- Counting One-to-One functions:
- How many One-to-One functions are there from a set with *m* elements to a set with *n* elements?
- NYX 86 010 6228 3324 京P CA129 license plates

IP Address Example

Some facts about the Internet Protocol, version 4:

- Valid computer addresses are in one of 3 types:
  - A *class A* IP address contains a 7-bit "netid" ≠ 1^7, and a 24-bit "hostid"
  - A *class B* address has a 14-bit netid and a 16-bit hostid.
  - A *class C* addr. Has 21-bit netic and an 8-bit hostid.
- The 3 classes have distinct headers (0, 10, 110)
- Hostids that are all 0s or all 1s are not allowed.

How many valid computer addresses are there?

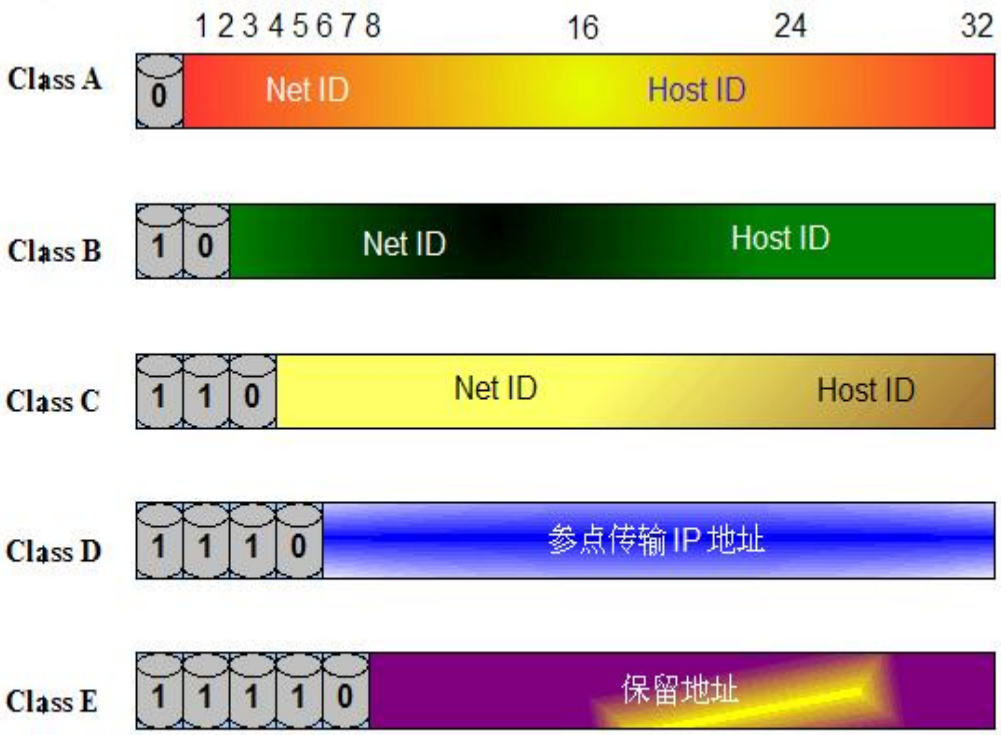


图 1-3 IP 地址类别

e.g., bupt.edu is 123.127.134.30

IP address solution

- (# addrs)
  = (# class A) + (# class B) + (# class C)
  (by sum rule)
- # class A = (# valid netids)·(# valid hostids)
  (by product rule)
- (# valid class A netids) = $2^7 - 1 = 127$.
- (# valid class A hostids) = $2^{24} - 2 = 16{,}777{,}214$.
- Continuing in this fashion we find the answer is:
  3,737,091,842 (3.7 billion IP addresses)

Inclusion-Exclusion Principle (§ §8.5)

- Suppose that $k \leq m$ of the ways of doing task 1 also simultaneously accomplish task 2.
  - And thus are also ways of doing task 2.
- Then, the number of ways to accomplish "Do either task 1 or task 2" is $m+n-k$.
- Set theory: If A and B are not disjoint, then $|A \cup B| = |A| + |B| - |A \cap B|$.
  - If they are disjoint, this simplifies to $|A|+|B|$.

Inclusion/Exclusion Example

- Some hypothetical rules for passwords:
  - Passwords must be 2 characters long.
  - Each character must be a letter a-z, a digit 0-9, or one of the 10 punctuation characters !@#$%^&*().
  - Each password must contain at least 1 digit or punctuation character.

Setup of Problem

- A legal password has a digit or punctuation character in position 1 **or** position 2.
  - These cases overlap, so the principle applies.
- (# of passwords w. OK symbol in position #1) = (10+10)·(10+10+26)
- (# w. OK sym. in pos. #2): also 20·46
- (# w. OK sym both places): 20·20
- Answer: 920+920−400 = 1,440

The Division Rule

- There are *n/d* ways to do a task if it can be done using a procedure that can be carried out in *n* ways, and for every way *w*, exactly *d* of the *n* ways correspond to way w.

Example

- How many different ways are there to seat four people around a circular table, where two seatings are considered the same when each person has the same left neighbor and the same right neighbor?

- Solution: We arbitrarily select a seat at the table and label it seat 1. We number the rest of the seats in numerical order, proceeding clockwise around the table. Note that are four ways to select the person for seat 1, three ways to select the person for seat 2, two ways to select the person for seat 3, and one way to select the person for seat 4. Thus, there are 4! = 24 ways to order the given four people for these seats.

- However, each of the four choices for seat 1 leads to the same arrangement, as we distinguish two arrangements only when one of the people has a different immediate left or immediate right neighbor. Because there are four ways to choose the person for seat 1, by the division rule there are 24/4 = 6 different seating arrangements of four people around the circular table.

Pigeonhole Principle (§ 6.2)

- A.k.a. the "Dirichlet drawer principle"
- If $\geq k+1$ objects are assigned to k places, then at least 1 place must be assigned ≥2 objects.
- In terms of the assignment function:
  - If $f{:}A \rightarrow B$ and $|A| \geq |B|+1$, then some element of B has ≥2 preimages under f.
    - I.e., f is not one-to-one.

Example of Pigeonhole Principle

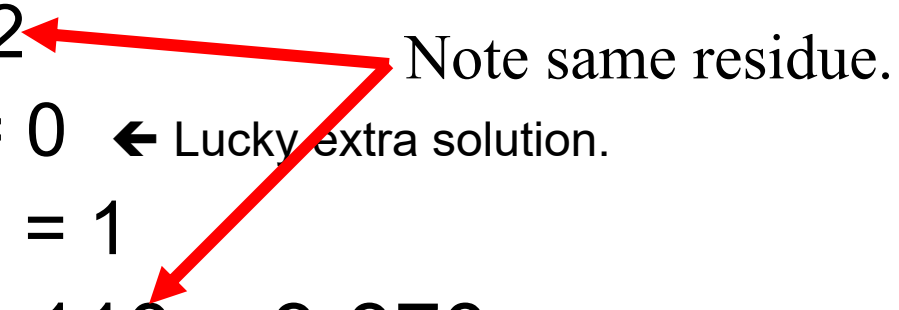
- There are 101 possible numeric grades (0%-100%) rounded to the nearest integer.
  - Also, there are >101 students in this class.
- Therefore, there must be at least one (rounded) grade that will be shared by at least 2 students at the end of the semester.
  - I.e., the function from students to rounded grades is *not* a one-to-one function.

Fun Pigeonhole Proof (Ex. 4, p.401)

- **Theorem:** $\forall n \in \mathbf{N}$, $\exists$ a multiple $m>0$ of n m has only 0's and 1's in its decimal expansion!
- **Proof:** Consider the n+1 decimal integers 1, 11, 111, …, $\underbrace{1\cdots1}_{n+1}$. They have only n possible residues mod n. So, take the difference of two that have the same residue. The result is the answer! □

A Specific Case

- Let n=3. Consider 1, 11, 111, 1111.
  - 1 mod 3 = 1
  - 11 mod 3 = 2
  - 111 mod 3 = 0 ← Lucky extra solution.
  - 1,111 mod 3 = 1



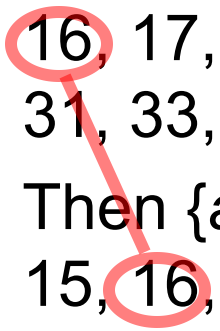
- 1,111 − 1 = 1,110 = 3·370.
  - It has only 0's and 1's in its expansion.
  - Its residue mod 3 = 0, so it's a multiple of 3.

Another Fun Example

- Suppose that next June, the Marlins baseball team plays at least 1 game a day, but ≤45 games total. Show there must be some sequence of consecutive days in June during which they play *exactly* 14 games.
  - **Proof:** Let a_j be the number of games played on or before day j. Then, $a_1,\ldots,a_{30} \in \mathbf{Z}^+$ is a sequence of 30 distinct integers with $1 \le a_j \le 45$. Therefore $a_1+14,\ldots,a_{30}+14$ is a sequence of 30 distinct integers with $15 \le a_j+14 \le 59$. Thus, $(a_1,\ldots,a_{30},a_1+14,\ldots,a_{30}+14)$ is a sequence of 60 integers from the set {1,..,59}. By the Pigeonhole Principle, two of them must be equal, but $a_i \neq a_j$ for $i \neq j$. So, $\exists ij$: $a_i = a_j+14$. Thus, 14 games were played on days $a_j+1, \ldots, a_i$.

Baseball problem illustrated

- Example of $\{a_i\}$: Note all elements are distinct.
  - 1, 2, 4, 5, 7, 8, 10, 11, 13, 14, 16, 17, 19, 21, 22, 23, 25, 27, 29, 30, 31, 33, 34, 36, 37, 39, 40, 41, 43, 45
  - Then $\{a_i+14\}$ is the following sequence 15, 16, 18, 19, 21, 22, 24, 25, 27, 28, 30, 32, 33, 35, 36, 37, 39, 41, 43, 44, 45, 47, 48, 50, 51, 53, 54, 55, 57, 59



Thus, for example, exactly 14 games were played during days 3 to 11:
2+1+2+1+2+1+2+1+2

- In any 60 integers from 1-59 there must be some duplicates, indeed we find the following ones:
  - 16, 19, 21, 22, 25, 27, 30, 33, 36, 37, 39, 41, 43, 45

Generalized Pigeonhole Principle

- If N objects are assigned to k places, then at least one place must be assigned at least $\lceil N/k \rceil$ objects.
- *E.g.*, there are N=280 students in this class. There are k=52 weeks in the year.
  - Therefore, there must be at least 1 week during which at least $\lceil 280/52 \rceil = \lceil 5.38 \rceil = 6$ students in the class have a birthday.

Proof of G.P.P.

- By contradiction. Suppose every place has $< \lceil N/k \rceil$ objects, thus $\leq \lceil N/k \rceil - 1$.
- Then the total number of objects is at most

$$k\left(\left\lceil \frac{N}{k} \right\rceil - 1\right) < k\left(\left(\frac{N}{k} + 1\right) - 1\right) = k\left(\frac{N}{k}\right) = N$$

- So, there are less than N objects, which contradicts our assumption of N objects!

□

G.P.P. Example

- Given: There are 280 students in the class.
  - Without knowing anybody's birthday, what is the largest value of *n* for which we can prove using the G.P.P. that at least *n* students must have been born in the same month?
- Answer:

$$\lceil 280/12 \rceil = \lceil 23.3 \rceil = 24$$

Example 7

- A) How many cards must be selected from a standard deck of 52 cards to guarantee that at least three cards of the same suit are chosen?
- B) How many must be selected to guarantee that at least three hearts are selected?

Some elegant application

- Example.11 Show that among any n+1 positive integers not exceeding 2n there must be an integer that divides one of the other integers.
- Example 13 Assume that in a group of six people, each pair of individuals consists of two friends or two enemies. Show that there are either three mutual friends or three mutual enemies in the group.
- Ramsy number

Solution: Write each of the $n + 1$ integers a_1, $a_2, \ldots, a_{n+1}$ as a power of 2 times an odd integer.In other words, let $a_j = 2^{kj}q_j$ for $j = 1, 2, \ldots, n + 1$, where kj is a nonnegative integer and q_j is odd. The integers $q_1, q_2, \ldots, q_{n+1}$ are all odd positive integers less than $2n$. Because there are only n odd positive integers less than $2n$, it follows from the pigeonhole principle that two of the integers $q_1, q_2, \ldots, q_{n+1}$ must be equal. Therefore, there are distinct integers i and j such that $q_i = q_j$. Let q be the common value of q_i and q_j. Then, $a_i = 2^{ki}q$ and $a_j = 2^{kj}q$. It follows that if $ki < kj$, then a_i divides a_j; while if $ki > kj$, then a_j divides a_i .

Some elegant application

- Example 13 Assume that in a group of six people, each pair of individuals consists of two friends or two enemies. Show that there are either three mutual friends or three mutual enemies in the group.
- Ramsy number

- *Solution:* Let *A* be one of the six people. Of the five other people in the group, there are either three or more who are friends of *A*, or three or more who are enemies of *A*. This follows from the generalized pigeonhole principle, because when five objects are divided into two sets, one of the sets has at least $\lceil 5/2 \rceil = 3$ elements. In the former case, suppose that *B*, *C*, and *D* are friends of *A*. If any two of these three individuals are friends, then these two and *A* form a group of three mutual friends. Otherwise, *B*, *C*, and *D* form a set of three mutual enemies. The proof in the latter case, when there are three or more enemies of *A*,

Theorem 3

- Every sequence of n^2+1 distinct real numbers contains a subsequence of length n+1 that is either strictly increasing or strictly decreasing.
- 8, 11, 9, 1, 4 ,6, 12, 10, 5, 7
- 1 4 6 10

Permutations (§ 6.3)

- A *permutation* of a set *S* of objects is a sequence that contains each object in *S* exactly once.
  - An ordered arrangement of *r* distinct elements of *S* is called an *r-permutation of S*.
- The number of *r*-permutations of a set with $n=|S|$ elements is

$$P(n,r) = n(n-1)\ldots(n-r+1) = n!/(n-r)!$$

Permutation Example

- A terrorist has planted an armed nuclear bomb in your city, and it is your job to disable it by cutting wires to the trigger device. There are 10 wires to the device. If you cut exactly the right three wires, in exactly the right order, you will disable the bomb, otherwise it will explode! If the wires all look the same, what are your chances of survival?

$P(10,3) = 10 \cdot 9 \cdot 8 = 720,$
so there is a 1 in 720 chance that you'll survive!

Combinations (§ 6.3)

- An *r*-combination of elements of a set *S* is simply a subset $T \subseteq S$ with *r* members, $|T|=r$.
- The number of *r*-combinations of a set with $n=|S|$ elements is

$$C(n,r) = \binom{n}{r} = \frac{P(n,r)}{P(r,r)} = \frac{n!/(n-r)!}{r!} = \frac{n!}{r!(n-r)!}$$

- Note that $C(n,r) = C(n, n-r)$
  - Because choosing the *r* members of *T* is the same thing as choosing the $n-r$ non-members of *T*.

Combination Example

- How many distinct 7-card hands can be drawn from a standard 52-card deck?
  - The order of cards in a hand doesn't matter.
- Answer $C(52,7) = P(52,7)/P(7,7)$
  $= 52 \cdot 51 \cdot 50 \cdot 49 \cdot 48 \cdot 47 \cdot 46 \,/\, 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$

$17 \quad 10 \quad 7 \quad 8 \quad 2$

$52 \cdot 17 \cdot 10 \cdot 7 \cdot 47 \cdot 46 = 133{,}784{,}560$

Example 14

- How many bit string of length n exactly r 1s?

- C(n,r)

Example 15

- Suppose that there are 9 faculty members in the mathematics department and 11 in the computer science department. How many ways are there to select a committee to develop a discrete mathematics course at a school if the committee is to consist of three faculty members from the mathematics department and four from the computer science department?

Skipping some sections

- Rosen 6rd edition…
- § 4.4, Binomial Coefficients
  - Useful in algebra… $C(n,r)$ occurs as a coefficient of terms in polynomials like $(a+b)^n$.
  - Pascal's identity and Vandermonde's identity.
- § 4.6, Generating Permutations & Combinations

Binomial Coefficients

- Goal: Try to find the coefficients of the expansion of powers of binomial expressions, e.g., $(x+y)^n$, without actually expanding the expression into multiplications of the sum.

Binomial Coefficients

- Binomial Theorem:

$$(x+y)^n = \sum_{j=0}^{n} \binom{n}{j} x^{n-j} y^j$$

- Ex 1: What's the expansion of $(x+y)^4$?
- Ex 2: What's the coefficient of $x^{12}y^{13}$ in the expansion of $(x+y)^{25}$?
- Ex 3: What's the coefficient of $x^{12}y^{13}$ in the expansion of $(2x-3y)^{25}$?

Binomial Coefficients

- Corollary 1:

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n$$

- Corollary 2:

$$\sum_{k=0}^{n} (-1)^k \binom{n}{k} = 0$$

- Corollary 3:

$$\sum_{k=0}^{n} 2^k \binom{n}{k} = 3^n$$

Pascal's Identity and Triangle

- Pascal's Identity: Let n>0, k>0, n≥k

$$\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$$

- Pascal's Triangle: (refers to the book)

6.5Generalized Permutations and Combinations

- Deals with how to solve counting problems where elements may be used more than once. → Permutation/Combination with repetition problems.
- Deals with how to solver counting problems with in which some elements are indistinguishable. → Permutations with indistinguishable-object problems.

Permutation with repetition

- Theorem 1: The number of *r*-permutation of a set of *n* objects with repetition allowed is n^r.
- Ex 1: How many strings of length *n* can be formed from the English alphabet?
- Answer: 26^n

Combination with repetition

- Theorem 2: There are C(n+r-1, r) r-combinations from a set with n elements when repetition of elements is allowed.
- Ex 1: How many ways are there to select four pieces of fruit from a bowl containing apples, oranges, and pears if the order in which the pieces are selected does not matter, only the type of fruit and not the individual piece matters, and there are at least four pieces of each type of fruit in the bowl?
- Answer: $r = 4$, $n = 3$ → = C(3+4-1, 4) = C(6, 4) = 15

Solution: To solve this problem we list all the ways possible to select the fruit. There are 15 ways:

4 apples； 4 oranges； 4 pears；

3 apples, 1 orange； 3 apples, 1 pear； 3 oranges, 1 apple； 3 oranges, 1 pear； 3 pears, 1 apple； 3 pears, 1 orange；

2 apples, 2 oranges； 2 apples, 2 pears； 2 oranges, 2 pears；

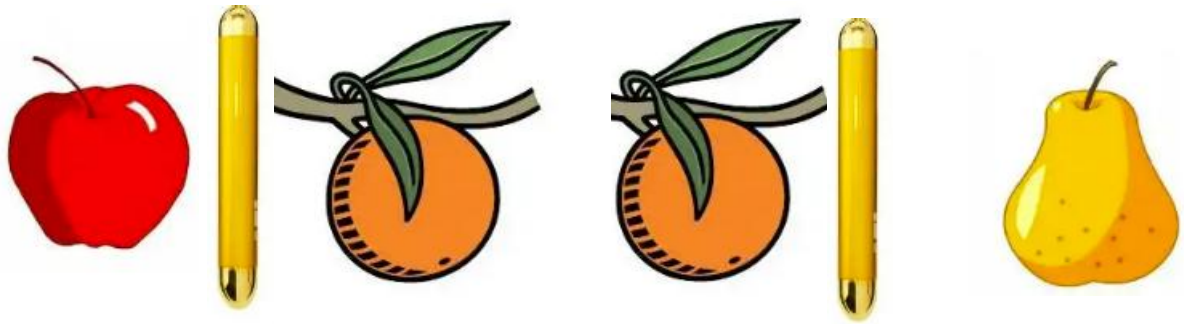
2 apples, 1 orange, 1 pear； 2 oranges, 1 apple, 1 pear； 2 pears, 1 apple, 1 orange

The solution is the number of 4-combinations with repetition allowed from a three-element set,{*apple*, *orange*, *pear*}.



2 apples, 1 orange, 1 pear

C(6,4)



1 apples, 2 orange, 1 pear

Combination with repetition

- Ex 2: How many ways are there to select five bills from a cash box containing $1 bills, $2 bills, $5 bills, $10 bills, $20 bills, $50 bills, and $100 bills? Assume that the order in which the bills are chosen does not matter, that the bills of each denomination are indistinguishable, and that there are at least five bills of each type.
- Answer: $r = 5$, $n = 7$ → = C(7+5-1, 5) = C(11, 5)

= 462

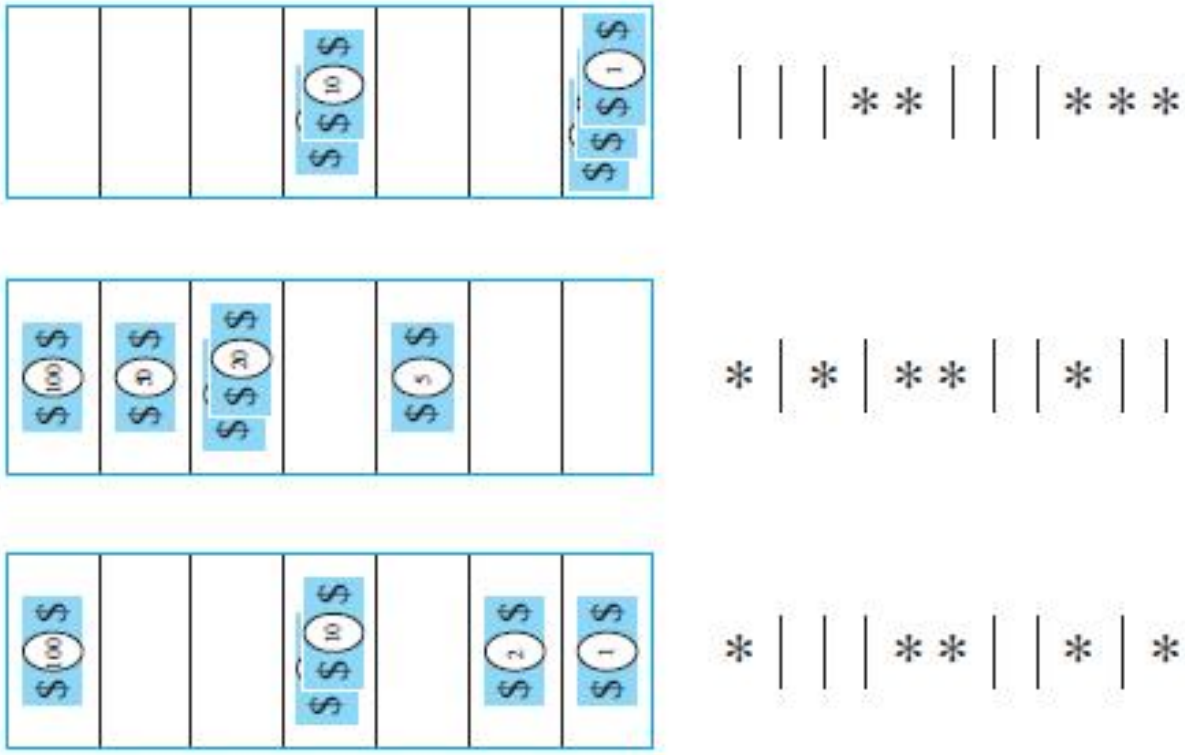


FIGURE 2 Examples of ways to select five bills.

Example 5

- How many solutions does the equation
- $X_1+X_2+X_3=11$
- Have, where X_1,X_2,and X_3 are nonnegative integers?

$C(3 + 11 - 1, 11) = C(13, 11) = C(13, 2) = 13 \cdot 12/1 \cdot 2 = 78$

Example 6

What is the value of k after the following pseudocode has been executed?

- *k:=0*
- *for i_1:=1 to n*
- *for i_2:=1 to i_1*
- *.*
- *.*
- *for i_m:=1 to i_{m-1}*
- *k:=k+1*

- *k:=0*
- *for i_1:=1 to 3*
- *for i_2:=1 to i_1*
- *for i_3:=1 to i_2*
- *for i_4:=1 to i_3*
- *k:=k+1*

Combinations and Permutations with and without Repetition

Type	Repetition Allowed?	Formula
r-permutations	No	$\frac{n!}{(n-r)!}$
r-combinations	No	$\frac{n!}{r!(n-r)!}$
r-permutations	Yes	n^r
r-combinations	Yes	$\frac{(n+r-1)!}{r!(n-1)!}$

Permutations with Indistinguishable Objects

- Theorem 3: The number of different permutations of n elements, where there are n_1 indistinguishable objects of type 1, n_2 indistinguishable objects of type 2, …, and n_k indistinguishable objects of type k, is

$$\frac{n!}{n_1!n_2!\cdots n_k!}$$

Permutations with Indistinguishable Objects

- Example: How many different strings can be made by reordering the letters of the word SUCCESS?
- Answer: $n = 7$, n_1 (#Ss) = 3, n_2 (#Cs) = 2, n_3 (#Us) = 1, n_4 (#Es) = 1 →

$$= \frac{7!}{3!2!1!1!} = 420$$

n_1 n_2 n_3 n_4

Distributing Objects into Boxes

- Theorem 4: The number of ways to distribute n distinguishable objects into k distinguishable boxes so that n_i objects are placed into box i, $i = 1, 2, \ldots, k$, equals

$$\frac{n!}{n_1!n_2!\cdots n_k!}$$

Distributing Objects into Boxes

- Example: How many ways are there to distribute hands of 5 cards to each of four players from the standard deck of 52 cards?
- Answer: $n = 52$, $k = 5$ →

$$= \frac{52!}{5!5!5!5!32!}$$

4 players, 5 cards each The rest (52-20) cards

Indistinguishable objects and distinguishable boxes

- How many ways are there to place 10 indistinguishable balls into eight distinguishable bins?

- This means there are C(n+r-1,n-1) ways to place r indistinguishable objects into n distinguishable boxes.

distinguishable objects and indistinguishable boxes

- How many ways are there put four different employees into three indistinguishable offices, when each office can contain any number of employees?
- We can put all four employees into one office in exactly one way, represented by {{A,B,C,D}}. We can put three employees into one office and the fourth employee into a different office in exactly four ways, represented by {{A,B,C}, {D}}, {{A,B,D}, {C}},{{A,C,D}, {B}}, and {{B,C,D}, {A}}.

distinguishable objects and indistinguishable boxes

- We can put two employees into one office and two into a second office in exactly three ways, represented by {{A,B}, {C,D}}, {{A,C}, {B,D}}, and {{A,D}, {B,C}}. Finally, we can put two employees into one office, and one each into each of the remaining two offices in six ways, represented by {{A,B}, {C}, {D}}, {{A,C}, {B}, {D}},{{A,D}, {B}, {C}}, {{B,C}, {A}, {D}}, {{B,D}}, {A}, {C}}, and {{C,D}, {A}, {B}}.

- Counting all the possibilities, we find that there are 14 ways to put four different employees into three indistinguishable offices.
- Let $S(n, j)$ denote the number of ways to distribute n distinguishable objects into j indistinguishable boxes so that no box is empty. The numbers $S(n, j)$ are called **Stirling numbers of the second kind**.

Indistinguishable objects and indistinguishable boxes

- How many ways are there to pack six copies of the same book into four identical boxes, where a box can contain as many as six books?
- 6 5,1 4,2 4,1,1 3,3 3,2,1 3,1,1,1
- 2,2,2 2,2,1,1
- Nine ways

$p_k(n)$

- If $a_1+a_2+\cdots+a_j = n$, where $a_1, a_2, \ldots, a_j$ are positive integers with $a_1 \geq a_2 \geq \cdots \geq a_j$, we say that
- $a_1, a_2, \ldots, a_j$ is a **partition** of the positive integer n into j positive integers.
- if $p_k(n)$ is the number of partitions of n into at most k positive integers, then there are $p_k(n)$ ways to distribute n indistinguishable objects into k indistinguishable boxes.

§ 6.6, Generating Perms. & Combs.

- We will go over algorithms for:
  - Generating the next largest permutation, in lexicographic order.
  - Generating the next largest bit string.
    - Remember, a bit string can represent a combination.
  - Generating the next *r*-combination in lexicographic order.
- Also we'll give recursive algorithms for generating permutations, combinations, and *r*-combinations.

Generating All Permutations

procedure *genAllPerms*(*n*>0: integer)
{output all permutations of the integers 1,..,*n*, in order from smallest to largest}
for *i*:=1 to *n* **begin** $used_i$ = **F end**
{none of the integers have been used yet}
recursiveGenPerms(*i*,*n*)

The *recursiveGenPerms* procedure is on the next slide…

Recursive Permutation Generator

```
procedure recursiveGenPerms(i,n)
   if i>n then begin {We're done, print the answer.}
        for k := 1 to n print perm_k ; print newline
   end else
        for j := 1 to n {Consider all poss. next items.}
             if ¬used_j then begin {Choose item j.}
                  used_j := T;  perm_i := j
                  recursiveGenPerms(i+1,n)
                  used_j := F       {Now back up}
             end
```

Next Permutation in Order

- Given an existing permutation $a_1,\ldots,a_n$ of $\{1,\ldots,n\}$, how do we find the *next* one?
- Outline of procedure:
  - Find largest j such that $a_j < a_{j+1}$.
  - Find the smallest integer in $a_{j+1},\ldots,a_n$ that is greater than a_j. Put it in position j.
  - Sort the remaining integers in $a_j,\ldots,a_n$ from smallest to largest. Put them at j+1 through n.

Next-Permutation Procedure

procedure *nextPerm*($a_1,\ldots,a_n$: perm. {1,…,*n*})
j:=*n*−1; **while** $a_j > a_{j+1}$ and *j*>0 **do** *j*:= *j*−1
if *j*=0 **then return** {no more permutations}
k:=*n*; **while** $a_j > a_k$ **do** *k*:=*k*−1
swap(a_j, a_k); *r*:=*n*; *s*:=*j*+1
while *r*>*s* **do begin**
swap(a_r,a_s); *r*:=*r*−1; *s*:=*s*+1 **end**

Combination Generator

- Suppose we want to generate all combinations of the elements of the set {1, ..., *n*}.
  - Or any other set with *n* elements.
- A combination is just a subset.
  - And, a subset of *n* items can be specified using a bit-string of length *n*.
    - Each bit says whether the item is in the subset.
- Therefore, we can enumerate all combinations by enumerating all bit-strings of length *n*.

Recursive Bit-String Enumerator

procedure *recEnumBitStrings*(*soFar*,*n*)
  {enumerate all strings consisting of *soFar* concatenated with a bit-string of length *n*}
  if *n*=0 **then begin print** *soFar*; **return end**
  enumBitStrings(*soFar*·‘0’, *n*−1)
  enumBitStrings(*soFar*·‘1’, *n*−1)

procedure *enumBitStrings*(*n*∈**N**)
  recEnumBitStrings(ε, *n*) {*soFar*=empty str}

Generating the Next Bit String

- Note that this is essentially just a binary increment ("add 1") operation.

procedure *nextBitString*($b_{n-1}\ldots b_0$: bit string)
  i:=0; {start at right end of string}
  while b_i = 1 and $i<n$ **begin** {trailing 1s}
    b_i := 0; i:=i+1 **end** {change to 0s}
  if $i=n$ **then return** "no more" **else begin**
    b_i = 1; **return** $b_{n-1}\ldots b_0$ **end** {chg. 0→1}

Generating *r*-combinations

- How do we list all *r*-combinations of the set {1,…,*n*}?
- Since the order of elements in a combination doesn't matter, we can always list them from smallest to largest.
- We can thus do it by enumerating the possible smallest items, then for each, enumerating the possible next-smallest items, *etc.*

A Recursive *r*-comb. Generator

procedure *recEnumRCombs*(*min*,*j*,*r*,*n*)
  if *j*>*n* **then print** $comb_1,\ldots,comb_n$; **return**
  for *i*:=*min* **to** *n*−*r*+1 **begin**
    $comb_j = i$
    recEnumRCombs(*i*+1, *j*+1, *r*−1, *n*)
  end

Generating Next *r*-combination

procedure *nextRComb*($\{a_1,\ldots,a_r\}\subset\{1,\ldots,n\}$ with $a_i<a_{i+1}$)

{Find the last item that can be inc'd}
i:=r; **while** $a_i = n-r+i$ **do** i:=$i-1$
$a_i := a_i + 1$ {Increment it}
for $j := i+1$ **to** r {Set remaining items}
$a_j := a_j + j - i$ {to the subsequent #'s}
return $\{a_1,\ldots,a_r\}$

作业

- §6.1 48, 52
  §6.2 14, 28, 48
  §6.3 30, 46
  §6.5 14, 20, 36, 48